

A blue and red Formula E race car, number 8, is shown on a track. The car features the Venturi logo and various sponsor logos including ROHM, ZF, and Michelin. The driver is wearing a helmet with the Venturi logo.

ROHM
SEMICONDUCTOR

SiC – Accelerating the evolution of EV by performing 16,000 switching operations per second

Cultivating technology at the highest levels to support the future of Electric Vehicles

In Formula E motor sports, the world's premier racing class for electric vehicles, cars compete for victory by performing up to 16,000 switching operations per second. As an official technology partner, ROHM provides state-of-the-art Full-SiC power modules for the main inverter that comprises the core of the main drive system. This makes it possible to reduce switching loss during power conversion by as much as 75%*, dramatically reducing inverter size and weight. In Formula E cars, which rely solely on battery power, effectively utilizing energy by minimizing losses is key to victory.

Formula E is increasingly being used as a testing ground and platform for innovation by the world's leading automakers and electronics manufacturers, who are developing cutting-edge technology solutions that are expected to be adopted in cars of the future to achieve sustainable mobility.

ROHM is an official technology partner of Formula E team Venturi.

*Compared with Silicon-IGBT modules with the same current rating, at 150°C

Inverter Comparison

SiC achieves a significant reduction in both size and weight vs. silicon solutions



ROHM's SiC Power Device

LINEUP

Full-SiC Power Module

1200V
(80A / 120A / 300A /
400A / 600A)



SiC-MOSFET

1700V (3.7A~5.9A)
1200V (10A~95A)
650V (21A~118A)



SiC-SBD

1200V (5A~40A)
650V (2A~40A)



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